Hektor Buqaj

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Education

University of Michigan, School of Engineering Expected May 2027
B.S. in Data Science, GPA: **3.5** *Ann Arbor, MI*
Awards: University Honors (Winter 2024), Dean's List (Winter 2024).
Coursework: Machine Learning, Data Mining, Data Structures, Algorithms, Web Systems, Applied Linear Regression.

Technical Skills

Programming: Python, SQL, JavaScript/TypeScript, C++.
Web / APIs: React, Flask, REST APIs, Auth & Session Management.
Data / ML: Apache Spark, scikit-learn, PyTorch, Retrieval-Augmented Generation (RAG).
Cloud / DevOps: AWS (EC2), Docker, PostgreSQL/SQLite, CI/CD, Git.

Experience

Research Assistant: Pulmonary Vessel Segmentation Algorithm Jan 2026 – Present
University of Michigan Galban Lab *Ann Arbor, MI*
- Building and iterating on a chest CT preprocessing + training pipeline in Python/PyTorch for pulmonary vessel segmentation.
- Extending a published approach to differentiate arterial vs. venous structures; designing evaluation experiments and error analysis.
- Training and evaluating models on large medical imaging datasets using the Great Lakes high-performance computing cluster.
- Communicating results and documenting workflows to support reproducibility and cross-team collaboration.

Engineering Intern May 2025 – Aug 2025
JVIS USA, LLC *Shelby Township, MI*
- Analyzed engineering test datasets to identify trends, anomalies, and performance deviations to support troubleshooting decisions.
- Produced clear technical reports and data summaries for engineering and management stakeholders; translated findings into recommendations.
- Helped refine test procedures to improve measurement consistency and reliability while collaborating with engineers and technicians.

Projects

Movie Recommender | *React, Bootstrap, Flask, SQLite, scikit-learn* Sep 2025 – Present
- Developed a full-stack web application delivering personalized movie recommendations using collaborative filtering.
- Implemented secure authentication/session management and stored user rating history in SQLite for individualized experiences.
- Designed RESTful APIs to update recommendations as new ratings are submitted and prevent re-recommending previously rated titles.
- Improved responsiveness by optimizing database queries and API request/response patterns.

Power Outage Analysis | *Python, Regression, Statistical Modeling* Apr 2025
- Built end-to-end machine learning pipeline incorporating preprocessing, feature scaling, and regularization.
- Compared LASSO and Ridge regression to evaluate bias–variance tradeoffs on real-world outage data.
- Conducted statistical analysis to assess feature significance and model robustness.
- Translated analytical findings into clear insights for non-technical stakeholders.

GPT-2 From Scratch | *PyTorch, Distributed Training* Sep 2025 – Dec 2025
- Implemented the GPT-2 transformer architecture from first principles in PyTorch, including multi-head attention and feed-forward blocks.
- Built the training pipeline (tokenization, batching, optimization scheduling) and ran distributed workloads on the Great Lakes GPU cluster.
- Tuned learning rate, batch size, and regularization settings to improve training stability.

M-CLOUD UAV Software System | *ROS2, QGroundControl, UAV Systems* Aug 2025 – Present
- Developed radio interfaces and telemetry pipelines enabling long-distance UAV communication.
- Designed modular software architecture to support autonomous flight and mission extensibility.
- Collaborated with multidisciplinary engineering teams under formal aerospace design review constraints.

Leadership & Activities

Resident Advisor (Leadership & Crisis Management) | *University of Michigan* Aug 2024 – Dec 2024
- Supported an inclusive residential community for 100+ students through engagement, mentorship, and accountability.
- Mediated conflicts and responded to emergency/high-stress situations; coordinated with campus safety & professional staff.
- Managed on-call responsibilities while balancing academic workload in a fast-paced environment.